

Improving NPC Hint Relevance with a Player Knowledge Model in an Educational Detective Game

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Abstract

Educational games often use non-player characters (NPCs) to guide players, but a useful hint depends on what the player has already discovered and understood. Repeating information the player already knows can make guidance feel unnecessary, while giving a hint that does not fit the player's current investigation can reduce its relevance.

This study evaluates whether a unified Player Knowledge Model (PKM) can improve NPC hint selection in *Shadow Castle: STEM Detective*, an educational detective game built in Godot. Three hint-selection policies were compared using the same catalogue of eleven authored NPC hints: Condition A, a static evidence-based baseline; Condition B, a deterministic PKM-aware selector; and Condition C, an LLM-based selector using player state and the same hard-eligible hint catalogue.

The held-out evaluation contained 72 frozen player states. Forty-eight CORE scenarios were used for the primary confirmatory analysis, while twenty-four STRESS scenarios were reserved for secondary boundary testing. Human relevance labels were completed and frozen before any of the three conditions were evaluated.

On the CORE scenarios, RelevantHintRate was 32/48 (66.7%) for Condition A, 32/48 (66.7%) for Condition B, and 33/48 (68.8%) for Condition C. A and B had no discordant relevance outcomes. Comparisons of A versus C and B versus C each had only three discordant scenarios and an exact two-sided McNemar p-value of 1.000. Therefore, the experiment did not establish a statistically supported improvement in confirmatory hint relevance.

The PKM-aware deterministic selector did, however, avoid the two redundant teaching selections produced by the static baseline across all 72 scenarios. The LLM selector showed a larger descriptive difference in deciding when not to speak: it correctly selected SILENCE on seven scenarios where no human-valid hint existed and falsely abstained once. These results suggest that, under the tested catalogue and selector design, player-state modelling may be more useful for reducing unnecessary guidance and supporting appropriate abstention than for substantially increasing broad hint relevance.

1. Introduction

NPC dialogue is a common way to provide guidance in games. In an educational game, however, selecting a useful NPC response is more complicated than simply giving the next available clue. A player may already understand a concept, may already have investigated a location, or may be holding evidence that changes which explanation makes sense.

This creates a hint-selection problem. The game contains a fixed catalogue of authored lines, but a system must decide which line best matches the player's current state.

Shadow Castle: STEM Detective is an educational detective game in which players investigate rooms, collect evidence, complete science-related challenges, and question NPCs. The game includes concepts from chemistry, optics, and circuits. Because a player's investigation and knowledge change over time, the same NPC line may be useful in one state and unnecessary in another.

This project therefore asks whether representing player knowledge explicitly can improve NPC hint selection.

The study does not evaluate whether the game improves student learning. A player state marked **DEMONSTRATED** means only that the player passed the game's existing designated comprehension check. It does not establish long-term mastery, retention, first-attempt correctness, or unassisted learning.

2. Research Questions

Primary Research Question

Can a unified Player Knowledge Model improve NPC hint relevance and reduce redundant guidance in an educational detective game?

The comparison between Conditions A and B directly tests this question.

Secondary Research Question

Does an LLM-based hint selector provide meaningful advantages over deterministic adaptive rules?

The comparison between Conditions B and C is the primary comparison for this secondary question, with A versus C also reported.

3. Game and System Context

Shadow Castle: STEM Detective is a Godot-based educational detective game built around investigation and science challenges.

The study focuses on three NPCs:

- **Butler**, associated with the Chemistry Room
- **Gardener**, associated with the Greenhouse
- **Mechanic**, associated with the Circuit Room

The shared experimental hint catalogue contains eleven authored lines. All three conditions operate on this same catalogue. Condition C does not generate new player-facing dialogue; it selects an existing hint identifier or chooses SILENCE.

This design is important because it separates **hint selection** from **hint writing**. Differences between the conditions therefore come from how they choose among the available hints rather than from one condition receiving better-written content.

4. Player Knowledge Model

The Player Knowledge Model is a read-only representation derived from existing game state.

It tracks eight concepts:

1. indicator reaction
2. physical versus chemical change
3. spectrum
4. reflection
5. additive color
6. circuit continuity
7. circuit regulation
8. circuit fault isolation

Each concept can have one of four conceptual states:

- **UNSEEN**
- **LEARNING**
- **DEMONSTRATED**
- **ASSISTED**

In the current implementation, ASSISTED is reserved for future use because the game does not yet store a reliable assistance signal.

The PKM does not independently store new gameplay information. Instead, it derives its state from existing evidence and story progression.

Most importantly, **DEMONSTRATED does not mean mastery**. It means only that the player passed the designated in-game comprehension check.

5. Hint-Selection Conditions

5.1 Condition A — Static Evidence-Depth Baseline

Condition A does not use the PKM.

It selects hints according to a deterministic evidence hierarchy:

1. If the player holds the NPC's associated evidence, select the evidence-responsive hint.
2. If the player does not hold that evidence but holds other major evidence, select a contextual hint.
3. Otherwise, select the introductory fallback.

Condition A always returns a hint.

5.2 Condition B — Deterministic PKM-Aware Selector

Condition B uses the same catalogue and hard eligibility rules as Condition A but adds Player Knowledge Model information.

Its rules can prefer a hint when a related concept has already been demonstrated and can avoid certain redundant teaching selections.

Condition B is therefore designed as a refinement of Condition A rather than a completely separate system.

Like A, Condition B always returns a hint.

5.3 Condition C — LLM + PKM Selector

Condition C receives:

- the NPC
- room and progression information
- player evidence
- story state
- derived PKM information
- the hard-eligible hint candidates

It must return either one valid hint identifier or **SILENCE**.

The model used in the frozen evaluation was **claude-opus-5** through an isolated company-managed Claude Code transport.

Condition C never generated new player-facing dialogue.

The evaluation used one successful model invocation per scenario. No scenario was rerun because its answer appeared undesirable, and no model selection was manually corrected.

6. Experimental Design

The final held-out dataset, called heldout-v4, contained **72 frozen player states**.

The scenarios were divided into two groups:

CORE

48 scenarios, with 16 scenarios per NPC.

CORE was the primary confirmatory dataset.

STRESS

24 scenarios, with 8 scenarios per NPC.

STRESS was designed to contain boundary cases such as very early states, very late states, already-demonstrated concepts, and states in which abstaining might be useful.

STRESS is reported separately and is not used to replace the CORE conclusion.

The scenario-generation process considered more than 55,000 legal candidate states before selecting the final held-out set.

Previous held-out scenario fingerprints were excluded so that exact earlier states were not reused.

7. Human Annotation

Each scenario was reviewed to determine which hard-eligible hints were reasonable for the player's current state.

Annotations could contain:

- one valid hint
- multiple valid hints
- or **NONE**, meaning that none of the available hints was sufficiently appropriate

The annotation procedure deliberately allowed multiple correct answers because hint relevance is not always a single-answer classification problem.

Each annotation also received an ambiguity level:

- LOW
- MEDIUM
- HIGH

The final distribution was:

- 62 LOW
- 8 MEDIUM
- 2 HIGH

The two HIGH scenarios occurred because the game state did not record Gardener interaction history, meaning that whether the player had already heard a particular line could not be determined.

Human annotations were frozen before any v4 A, B, or C output was observed.

The annotation process was researcher-led. AI tools were used to prepare neutral scenario summaries. One disclosed exception, Scenario 72, used an assistant-proposed adjudication based on a decision rule the researcher had already established and was accepted into the frozen annotation set.

8. Metrics

8.1 RelevantHintRate

The primary metric was:

RelevantHintRate = relevant selected hints / all attempted scenarios

A delivered hint counted as relevant when its identifier appeared in the human VALID_HINTS set.

Under this frozen definition, SILENCE always counts as a relevance failure, even when the human annotation is NONE.

This rule was preserved from the earlier protocol rather than changed after the results were known.

8.2 AppropriateActionRate

A prospective secondary metric was used to represent whether the selector took the appropriate overall action.

An action counted as correct when:

- the system delivered a hint contained in VALID_HINTS, or
- the system selected SILENCE when VALID_HINTS was empty

This allows a system to receive credit for correctly choosing not to speak.

8.3 Additional Metrics

Additional measurements included:

- CorrectSilenceRate
- Coverage
- StateViolationRate
- RedundantHintRate
- RedundantWhenTeachable

Relevance and redundancy were intentionally treated as different properties.

A hint could still be reasonable in context while repeating a concept that had already been demonstrated.

9. Statistical Analysis

The primary statistical analysis used the 48 CORE scenarios.

Exact paired two-sided McNemar tests compared:

- A versus B
- A versus C
- B versus C

Discordant counts were reported alongside each p-value.

The preregistered analysis also specified Holm-adjusted values across the planned comparison family. The exact unadjusted p-values remained the primary statistical reporting.

Because the observed number of discordant CORE pairs was extremely small, the statistical comparisons had limited inferential resolution.

No composite score combining the different metrics was created.

10. Results

10.1 CORE Results

The primary confirmatory results were:

Metric	Condition A	Condition B	Condition C
RelevantHintRate	32/48 (66.7%)	32/48 (66.7%)	33/48 (68.8%)
AppropriateActionRate	32/48 (66.7%)	32/48 (66.7%)	35/48 (72.9%)
CorrectSilenceRate	0/7	0/7	2/7
Coverage	48/48	48/48	45/48
RedundantHintRate	1/48	0/48	0/45 delivered
State Violations	0	0	0

Conditions A and B had identical RelevantHintRate outcomes across the 48 CORE scenarios.

For A versus B, there were no discordant relevance outcomes, so an exact McNemar p-value was not defined.

For A versus C and B versus C:

- $b = 1$
- $c = 2$
- total discordant pairs = 3
- exact two-sided $p = 1.000$

Condition C therefore showed a small observed increase of one relevant scenario, but the experiment did not provide statistical support for a confirmatory relevance improvement.

The CORE AppropriateActionRate was 32/48 for A, 32/48 for B, and 35/48 for C.

For A versus C and B versus C on this secondary metric:

- $b = 1$
- $c = 4$
- $p = 0.375$

This difference was also not statistically significant.

10.2 STRESS Results

STRESS results are secondary robustness evidence rather than confirmatory results.

Metric	Condition A	Condition B	Condition C
RelevantHintRate	12/24 (50.0%)	12/24 (50.0%)	16/24 (66.7%)
AppropriateActionRate	12/24 (50.0%)	12/24 (50.0%)	21/24 (87.5%)
CorrectSilenceRate	0/8	0/8	5/8
Coverage	24/24	24/24	19/24
RedundantHintRate	1/24	0/24	0/19 delivered

For STRESS RelevantHintRate, A versus C and B versus C each had:

- $b = 0$
- $c = 4$
- exact $p = 0.125$

These comparisons remain low-information and secondary.

No frozen inferential test was performed for STRESS AppropriateActionRate, so the 21/24 versus 12/24 difference is reported descriptively rather than supplemented with a post-hoc test.

11. Redundancy Analysis

Across all 72 scenarios, Condition A produced two redundant teaching hints.

Condition B produced zero.

Condition C produced zero redundant hints among its 64 delivered responses.

The two A-versus-B action differences occurred in Scenarios 29 and 59.

In both scenarios:

- Condition A selected `h_gardener_leaf_colour`
- Condition B selected a non-redundant alternative
- both selections were nevertheless labelled human-relevant

This demonstrates why relevance and redundancy measure different properties.

However, the redundancy result is very sparse.

Only two redundant events occurred in the entire evaluation, and the explicit redundancy fallback rule in Condition B was not actually the rule responsible for these two differences. The differences arose through the selector's soft-preference behavior.

Therefore, the study does not establish that the explicit redundancy mechanism itself has been validated.

12. Abstention and SILENCE

Condition C selected SILENCE eight times.

Seven were frozen NONE scenarios, meaning that no available hint had been labelled human-valid.

One was a false abstention: Scenario 15 contained a valid Butler hint, but C remained silent.

This produced an important difference between the two main metrics.

Under `RelevantHintRate`, all eight SILENCE outputs were failures because the primary metric requires a delivered valid hint.

Under `AppropriateActionRate`, the seven correct NONE abstentions received credit.

Across all 72 scenarios:

- `A CorrectSilenceRate`: 0/15
- `B CorrectSilenceRate`: 0/15
- `C CorrectSilenceRate`: 7/15

This suggests that Condition C's most distinctive behavior was not a large increase in the relevance of delivered hints but its ability to decide when not to deliver one.

13. Disagreement Analysis

At least one of the three systems selected a different action in 38 of the 72 scenarios.

A and B differed in only two scenarios.

Most disagreement therefore came from Condition C.

Several patterns appeared in C's selections:

- C frequently preferred directional Butler testimony instead of generic Butler background.
- C selected the Gardener's reflection teaching line more frequently.
- C selected the Mechanic's series-circuit teaching line in states where A and B often selected investigative testimony.

These patterns should not automatically be interpreted as better behavior.

For the Mechanic, for example, C frequently selected different hints but achieved exactly the same relevance totals as A and B:

- CORE: 13/16 for all three
- STRESS: 6/8 for all three

Behavioral difference alone therefore did not imply improved relevance.

14. Discussion

14.1 Did the PKM Improve Hint Relevance?

The primary result does not support a broad relevance improvement from the tested deterministic PKM selector.

A and B both achieved:

32/48 = 66.7% RelevantHintRate

and had no discordant relevance outcomes on CORE.

This should not be interpreted as proof that the systems are equivalent. The study simply could not distinguish them on the primary confirmatory relevance outcome.

One reason is structural.

Condition B is a refinement of Condition A. In most scenarios, both systems return the same fallback selection.

A and B differed in action on only two of 72 scenarios, and both choices were human-valid in those scenarios. RelevantHintRate therefore had no opportunity to separate them.

14.2 Did the PKM Reduce Redundant Guidance?

A narrow difference was observed.

Condition A produced two redundant teaching selections across all 72 scenarios, while Condition B produced none.

This suggests that PKM-aware preferences can help avoid some unnecessary re-teaching.

However, the number of events is extremely small.

The study therefore supports only a limited observation about redundancy control, not a strong general conclusion.

14.3 Did the LLM Selector Improve Relevance?

Condition C achieved:

33/48 = 68.8%

compared with:

32/48 = 66.7%

for A and B.

This represents only one additional relevant CORE scenario.

The paired comparisons produced:

p = 1.000

with only three discordant pairs.

The study therefore did not establish a statistically supported confirmatory relevance improvement from the LLM selector.

14.4 Where Condition C Differed Most

Condition C showed a larger descriptive difference in AppropriateActionRate and in the STRESS scenarios.

Its strongest distinctive capability was abstention.

Unlike A and B, C could determine that none of the available hints were appropriate and return SILENCE.

It correctly did this seven times and incorrectly did it once.

This behavior was particularly visible in boundary states.

However, because STRESS was intentionally constructed from unusual boundary scenarios, these results should not be interpreted as representing the frequency of such situations during normal gameplay.

14.5 What the Results Suggest

Under this particular hint catalogue and selector design, player-state modelling appears more useful for:

- avoiding unnecessary re-teaching
- distinguishing between teaching and already-demonstrated concepts
- deciding whether speaking is appropriate at all

than for substantially increasing broad hint relevance.

This conclusion is limited to the tested game, hint catalogue, state representation, and policies.

15. Limitations

Several limitations constrain the conclusions.

First, the primary CORE dataset contained only 48 scenarios, and the actual paired systems produced very few discordant outcomes.

Second, the catalogue contained only eleven hints across three NPCs. Many scenarios contained only one or two eligible candidates, limiting the opportunity for different selection policies to produce meaningfully different answers.

Third, conversational history was not fully represented. The Butler had interaction-history state, but the Gardener and Mechanic did not. This caused two HIGH-ambiguity annotations.

Fourth, relevance annotations were researcher-led rather than produced by multiple independent annotators. No inter-rater agreement statistic was available.

Fifth, Scenario 72 included an assistant-proposed adjudication accepted by the researcher.

Sixth, Condition C is stochastic. Sampling controls such as temperature, top-p, top-k, and seed were not controllable through the evaluation transport. The frozen outputs are auditable, but exact regeneration by rerunning the model is not guaranteed.

Seventh, the evaluation used only one LLM model and one successful sample per scenario.

Finally, this experiment did not measure learning outcomes or real player experience.

16. Future Work

Future work should preserve the current frozen evaluation rather than modifying it after seeing the results.

Important next steps include:

- adding explicit interaction-history state for every NPC
- expanding the hint catalogue with more genuinely competing hint types
- including more hints capable of becoming redundant
- using multiple independent human annotators
- measuring inter-rater agreement
- evaluating the system with actual players
- recording attempts, assistance, and timing before making learning claims
- conducting a separately preregistered replication with more scenarios
- testing multiple LLMs or multiple samples per scenario when studying stochastic robustness

A future experiment with a larger and more diverse hint catalogue may provide a stronger test of whether player knowledge models change relevance rather than only redundancy.

17. Conclusion

This study evaluated three NPC hint-selection policies in *Shadow Castle: STEM Detective* using a frozen 72-scenario held-out evaluation.

The unified Player Knowledge Model did not improve overall confirmatory hint relevance under the tested deterministic selector. Conditions A and B both achieved 32/48 RelevantHintRate on CORE.

The PKM-aware selector did avoid the small number of redundant teaching selections produced by the static baseline, although the evidence was limited to two events and did not validate the explicit redundancy fallback mechanism.

The LLM-based selector achieved 33/48 CORE RelevantHintRate, only one scenario above A and B. This difference was not statistically supported.

Its more distinctive behavior appeared in action appropriateness and abstention. Condition C correctly remained silent on seven scenarios where no human-valid hint existed and falsely abstained once.

Overall, the findings suggest that, under this catalogue and policy design, modelling player state may be more useful for controlling what should not be repeated and when an NPC should not speak than for substantially increasing broad hint relevance.

These findings do not establish that LLM selectors are generally superior to deterministic approaches, and they do not demonstrate improved learning outcomes.

18. AI Assistance and Researcher Contribution

AI tools played a substantial role in the engineering workflow for this project.

The researcher defined the research questions, game context, experimental goals, annotation criteria, and interpretation rules. The researcher also made the human relevance decisions and accepted the final frozen annotations.

Coding agents assisted substantially with implementation, auditing, test harnesses, scripts, documentation, data checking, and report drafting.

A common development workflow was:

researcher specifies requirements → AI agent implements → researcher evaluates behavior and results → researcher provides revised requirements

The project therefore should not be interpreted as one in which every source-code line was independently handwritten or manually audited by the researcher.

During human annotation, AI tools helped prepare neutral summaries of complex player states. The AI was instructed not to select the valid hints, rank candidates, or reveal selector outputs.

One disclosed exception is Scenario 72, where an assistant proposed an adjudication based on a decision rule already established by the researcher. The researcher accepted that label into the frozen ground truth.

Condition C's use of claude-opus-5 as an experimental selector is separate from AI assistance used to build and analyze the project.

No v4 Condition A, B, or C output was visible when the human annotations were frozen.

19. Reproducibility and Research Integrity

The project used staged freezing of scenarios, annotations, and evaluation outputs.

The final v4 scenario artifact was generated and frozen before annotation.

Human annotations were then completed and frozen before any A/B/C evaluation output was observed.

Only after this freeze were Conditions A, B, and C evaluated.

Condition C completed all 72 scenarios with exactly one successful invocation per scenario, with no transport failure, schema retry, or manual correction.

The frozen evaluation recorded:

- 72 scenarios
- 48 CORE scenarios
- 24 STRESS scenarios
- 72/72 Condition C completions

- zero state violations across all three conditions
- no post-evaluation modification of the frozen annotations

The complete repository also preserves commit hashes, tags, scenario hashes, annotation fingerprints, raw outputs, evaluation tables, run logs, and metric files so that the experimental sequence can be audited.